

strands from extending. These strands could then be ordered from the shortest to the longest so the base sequence could be read. This rapid, accurate way of sequencing enabled Sanger to map the first human genome, the DNA of cell mitochondria, in 1981, and led the way for sequencing all 3 billion base pairs of the entire human genome.

Sanger's sequencing method enabled fragments of DNA to be isolated, cloned, and reattached to create modified genes to treat—or be resistant to—genetic disorders.

JOHN SULSTON

English biologist John Sulston was the first director of the Wellcome Sanger Institute genome research center and shared the 2002 Nobel Prize for work on programmed cell death.



The most important research carried out by Sulston (1942–2018) centered on *Caenorhabditis elegans*, a 0.03-in (1-mm) nematode worm. This worm has an exact number of cells, so it is ideal for studying cell differentiation, division, and in some cases programmed death. Sulston pushed for the genetic sequencing of *C. elegans*, which in 1988 became the first animal to have its entire genome sequenced. This led directly to the Human Genome Project (1990–2003), in which Sulston was a key contributor.

